25901167 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCOM201 - QUANTITATIVE TECHNIQUES FOR BUSINESS DECISIONS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions (Word/Phrase/Sentence)

Answer any **12** questions

Each question carries **1** marks

1. When the amount of change in one variable leads to a constant ratio of change in the other variable, correlation is said to be ---------- [U] / [CO1]
2. What is the maximum possible value of a regression coefficient? [U] / [CO2]
3. Which Correlation method is most suitable for studying the relationship between qualitative data? [U] / [CO1]
4. Expand SEE in regression analysis [U] / [CO2]
5. What is line of best fit? [U] / [CO3]
6. When variations increase along with trend, which model is preferred? [U] / [CO3]
7. State the multiplication theorem of probability if events are independent. [U] / [CO4]
8. How many different words can be formed out of the letters of the word "SCHOOL"? [A] / [CO4]
9. A logistics company tracks the primary reason for delivery delays. Traffic congestion accounts for 45% of delays, vehicle breakdowns account for 12%, and severe weather accounts for 18%. Assuming a delay is only ever logged under a [A] / [CO4]

single primary cause, what is the probability that a delayed shipment was held up by traffic, a breakdown, or weather?

10. Name the four approaches to the concept of probability. [U] / [CO4]

11. Why is Regression called a predictive tool? [U] / [CO2]

12. What is irregular variation? [U] / [CO3]

13. In how many ways can 3 red balls be selected from 9 red balls? [A] / [CO4]

14. Which theorem is used to calculate inverse probability? [U] / [CO4]

[1x12 = 12]

Part B

Short Answer/ Problem Type Questions

Answer any **4** questions

Each question carries **7** marks

15. Explain the importance of Time Series Analysis in decision making. [U] / [CO3]

16. Explain the meaning and significance of the concept of correlation? Does it always signify cause and effect relationship between two variables? How is the coefficient of correlation interpreted? [U] / [CO1]

17. Fit a trend line by the method of semi - average to the data given below: [A] / [CO3]

Year	Sales (₹)
2018	50
2019	70
2020	60
2021	80
2022	70
2023	100
2024	120
2025	90

18. The odds in favour of A solving a puzzle is 6 to 9 and against B solving it is 12 to 10. what is the probability that the puzzle is solved? [A] / [CO4]

19. A portfolio manager is selecting 4 stocks to add to a high-growth fund from a shortlist of 10 potential companies. The shortlist consists of: 3 Banking Stocks, 4 IT Stocks, 2 Pharma Stocks and 1 Auto Stock If the 4 stocks are selected at random, find the probability that: (i) The portfolio contains exactly one stock from each of the four sectors? (ii) The portfolio consists of the Auto stock and at least one Banking stock? [A] / [CO4]

20. Find Karl Pearsons Correlation co efficient from the following data [A] / [CO1]

Data:

X	1	3	5
Y	2	4	3

[7x4 = 28]

Part C

Essay/ Problem Type Questions

Answer any **2** questions

Each question carries **15** marks

21. Three judges rank ten finalists. Who shows the highest agreement? Justify your answer with rank correlation co efficient. [A] / [CO1]

Contestant	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Judge A	5	2	3	8	10	1	4	6	9	7
Judge B	8	5	1	2	9	4	7	3	10	6
Judge C	4	3	2	7	8	1	5	6	10	9

22. A panel of judges, A and B graded seven debaters and independently awarded the following marks: [A] / [CO2, CO5]

Debator	1	2	3	4	5	6	7
Marks by A	40	34	28	30	44	38	31
Marks by B	32	39	26	30	38	34	28

An eighth debator was awarded 36 marks by Judge A, while Judge B was not present. If Judge B was also present, how many marks would you expect him to award to the eighth debator assuming the same degree of relationship exists in judgement?

23. The annual profit (in ₹ crores) of MNO Ltd. is as follows: [A] / [CO3, CO5]

Year	2016	2017	2018	2019	2020	2021	2022
Profit	40	42	50	55	60	68	75

a) Fit a straight line trend by least squares method.

b) Tabulate the trend values.

c) Estimate the profit for 2023.

[15x2 = 30]